COOLANT CONTROL VALVE REMOVAL/INSTALLATION [SKYACTIV-D 1.5]

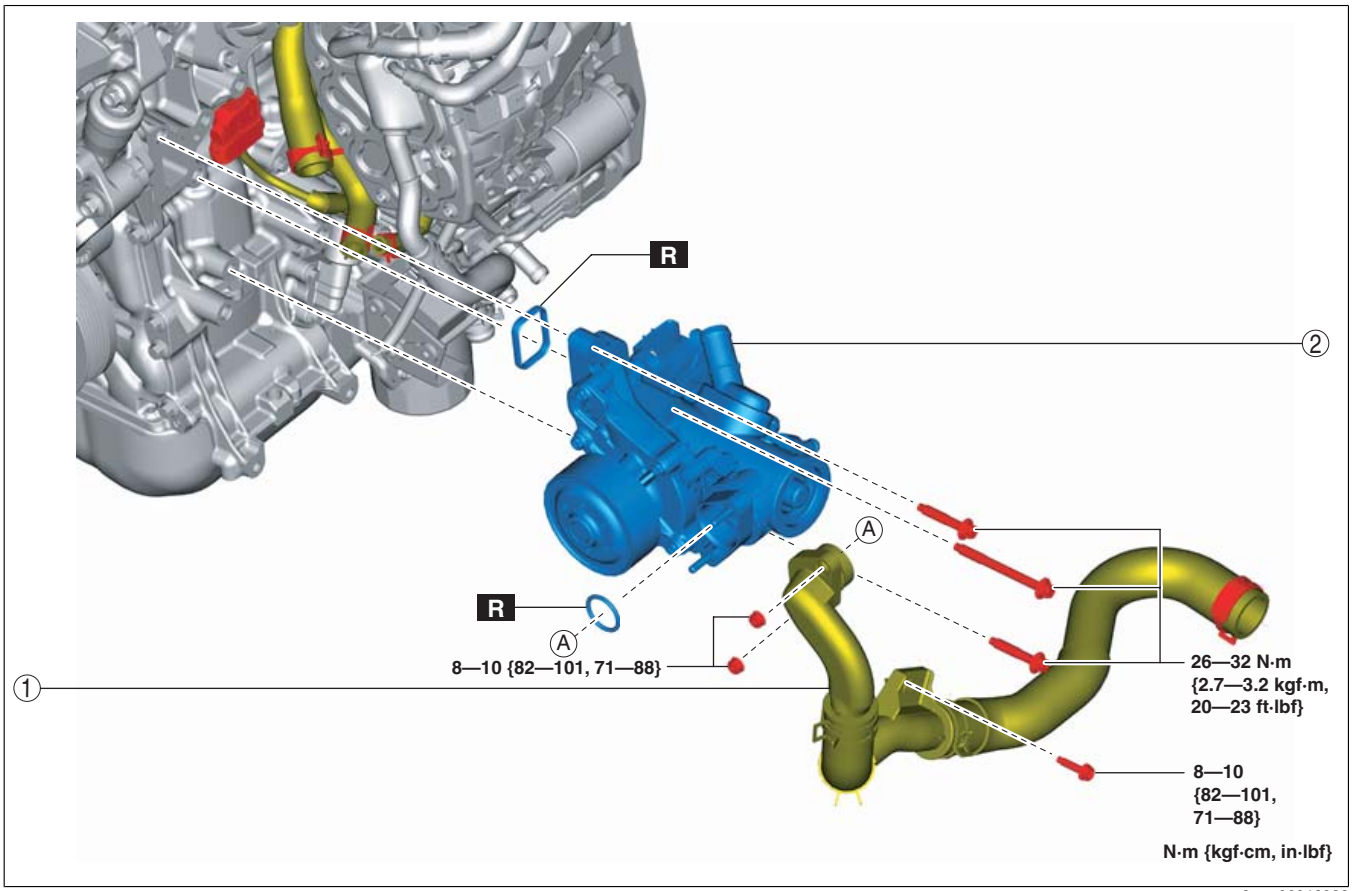
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Warning

- **If the cooling system cap or the water-cooled charge air cooler reserve tank cap is removed when the engine is hot, steam and boiling water may be injected, causing severe burns or severe injury. Perform the removal of the cooling system cap and water-cooled charge air cooler reserve tank cap when the engine is cool.**
- **Turn off the engine and wait until it is cool. Even then, be very careful when removing the cap. Wrap a thick cloth around it and slowly turn it counterclockwise to the first stop. Step back while the pressure escapes.**
- **When you are sure all the pressure is gone, press down on the cap using the cloth, turn it, and remove it.**

1. Turn the steering wheel completely to the right.
2. Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/ CONNECTION.)
3. Remove the front under cover No.1 and No.2. (See FRONT UNDER COVER No.1 REMOVAL/ INSTALLATION.) (See FRONT UNDER COVER No.2 REMOVAL/INSTALLATION.)
4. Drain the engine coolant. (See ENGINE COOLANT REPLACEMENT [SKYACTIV-D 1.5].)
5. Remove the front splash shield (RH). (See SPLASH SHIELD REMOVAL/INSTALLATION.)
6. Remove the drive belt. (See DRIVE BELT REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
7. Remove the engine cover. (See ENGINE COVER REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
8. Disconnect the service plug. (with i-ELOOP) (See SERVICE PLUG DISCONNECTION/CONNECTION [i-ELOOP].)
9. Remove the generator. (See GENERATOR REMOVAL/INSTALLATION [WITH i-ELOOP (SKYACTIV-D 1.5)].) (See GENERATOR REMOVAL/INSTALLATION [WITHOUT i-ELOOP (SKYACTIV-D 1.5)].)
10. Remove the generator bracket. (See ENGINE DISASSEMBLY/ASSEMBLY [SKYACTIV-D 1.5].)
11. Remove the coolant reserve tank. (See COOLANT RESERVE TANK REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
12. Remove in the order indicated in the table.
13. Install in the reverse order of removal.
14. Refill the engine coolant. (See ENGINE COOLANT REPLACEMENT [SKYACTIV-D 1.5].)
15. Inspect for engine coolant leakage. (See ENGINE COOLANT LEAKAGE INSPECTION [SKYACTIV-D 1.5].)

Step 1

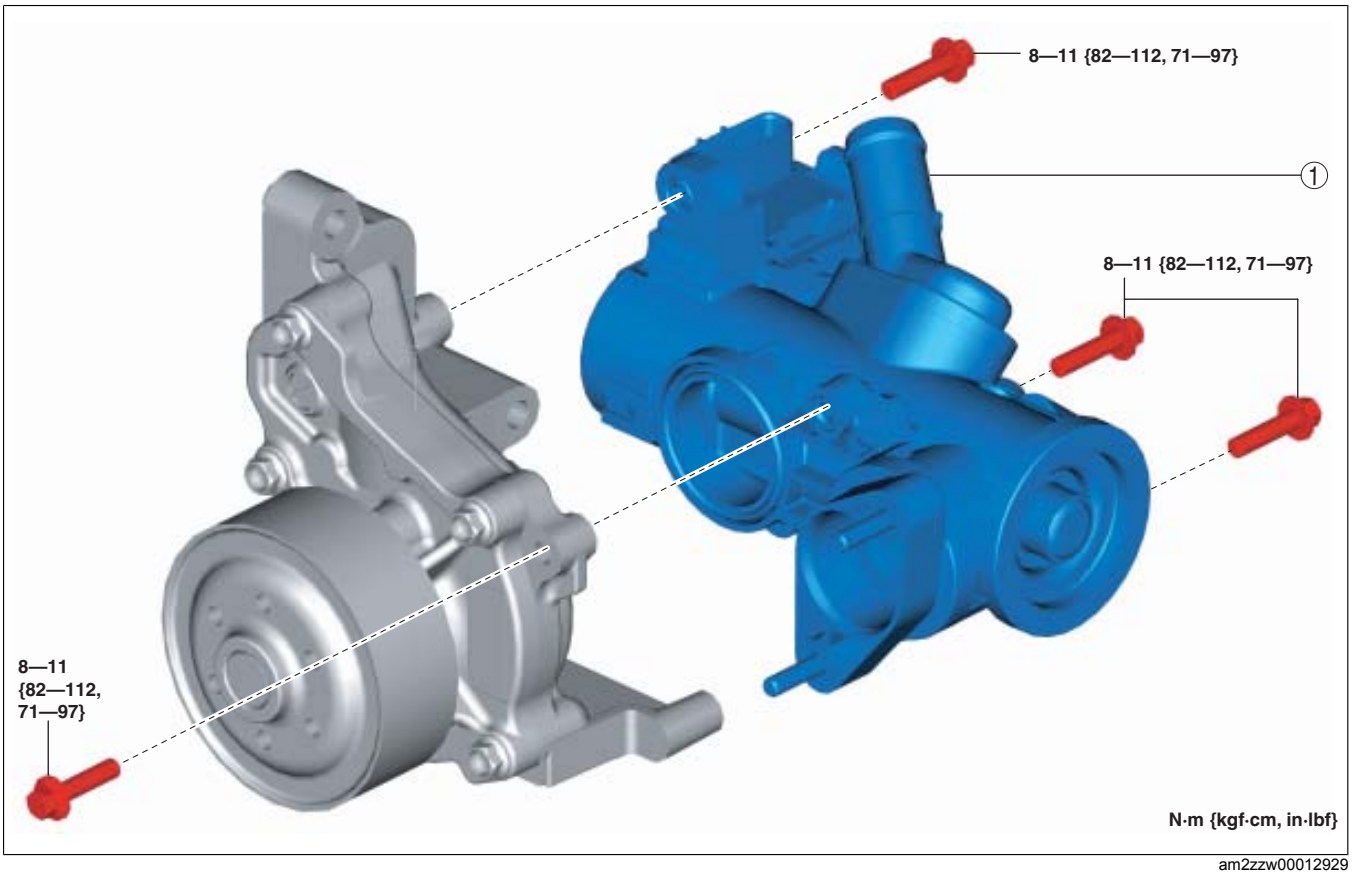


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1	Lower radiator hose component (See Lower Radiator Hose Component Removal Note.) (See Lower Radiator Hose Component Installation Note.)

2	Water pump component (See Water Pump Component Removal Note.) (See Water Pump Component Installation Note.)

Step 2

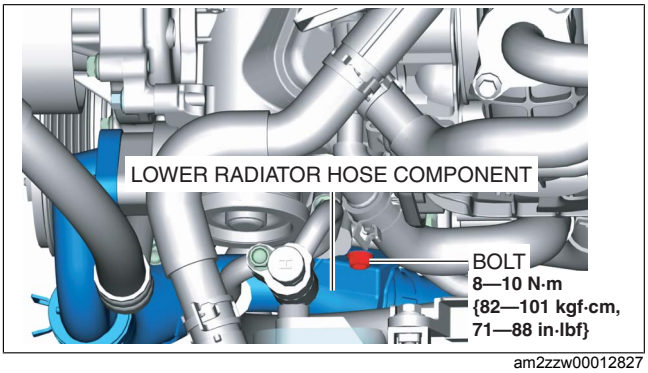


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1	Coolant control valve (See Coolant Control Valve Installation Note.)

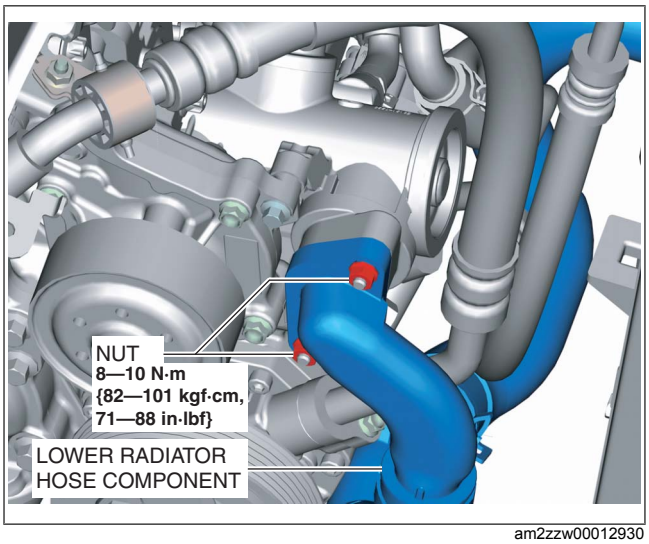
Lower Radiator Hose Component Removal Note

1. Remove the bolt shown in the figure.



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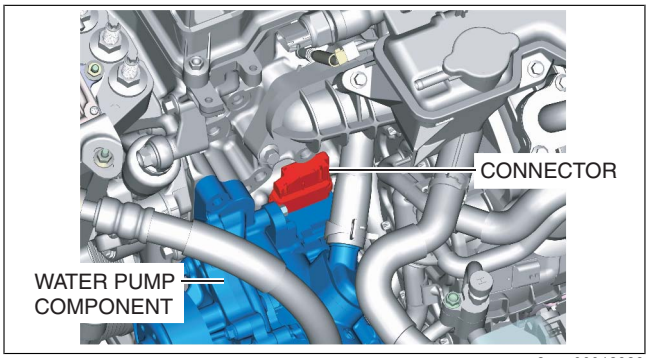
2. Remove the nuts shown in the figure.
3. Disconnect the lower radiator hose component (radiator side).
4. Remove the lower radiator hose component from the underside of the vehicle.



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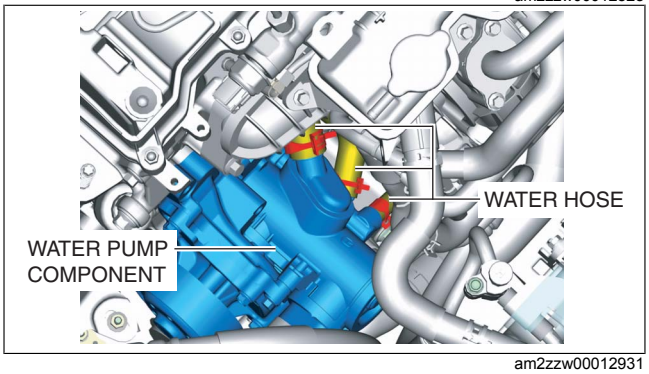
Water Pump Component Removal Note

1. Remove the A/C compressor with the pipes connected and secure the A/C compressor using wire or rope so that it is out of the way. (See A/C COMPRESSOR REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
2. Place an appropriate container under the engine to collect the draining engine coolant.
3. Remove the water pump component installation bolts and perform the following procedures while setting the water pump component aside.
   (1) Disconnect the connector.



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   (2) Disconnect the water hose.
4. Remove the water pump component.



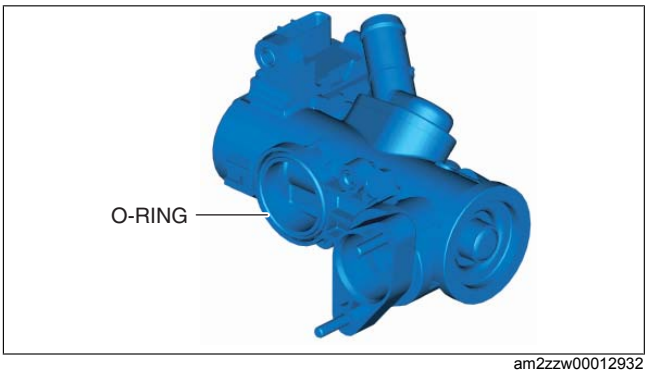
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Coolant Control Valve Installation Note

Caution

- **Do not apply oil (such as engine oil, ATF) to the O-ring. Otherwise, the O-ring could swell causing a seal malfunction.**

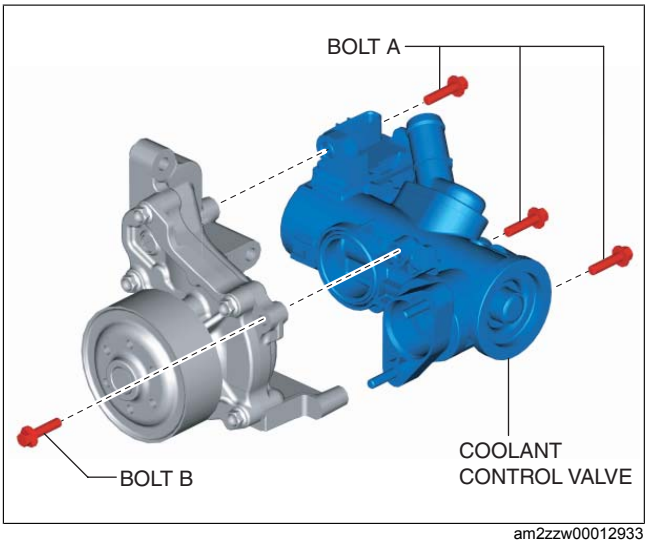
1. Visually inspect the engine coolant control valve O-ring.
   - If the O-ring is damaged, replace it.



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2. Install the coolant control valve using the following procedure.
   (1) Temporarily tighten bolts A shown in the figure.
   (2) Tighten the bolts to the specified torque in the order of B and A.

   Tightening torque
   8—11 N·m {82—112 kgf·cm, 71—97 in·lbf}

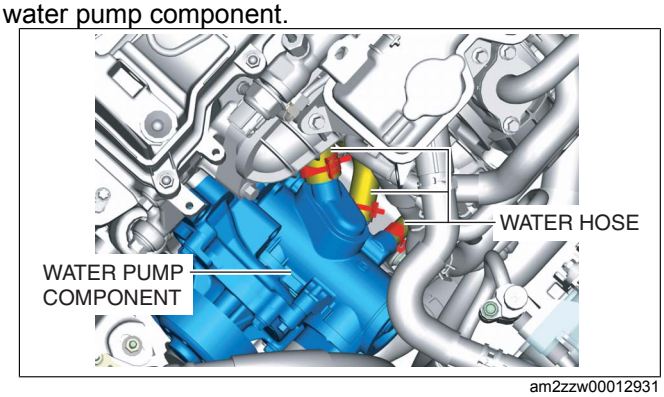


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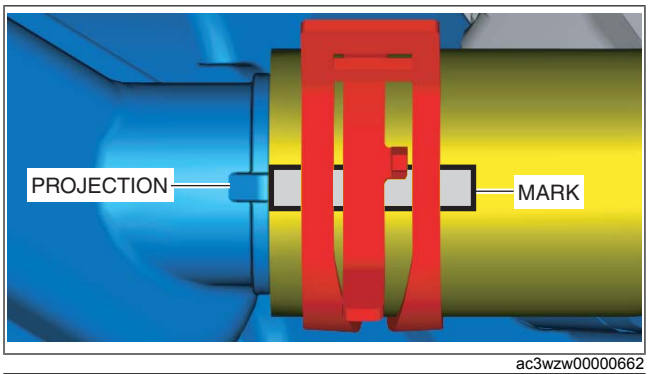
Water Pump Component Installation Note

Caution
- **Be aware of the following points, otherwise a poor seal could result.**
  - **— Do not allow oil (such as engine oil, ATF) to get on the gasket, otherwise the water pump gasket will swell.**
  - **— Verify that there is no lifting or detachment of the water pump gasket.**

1. Insert a new water pump gasket into the groove of the water pump component.
2. Connect the water hose to the water pump component so that the projection position is aligned with the mark.

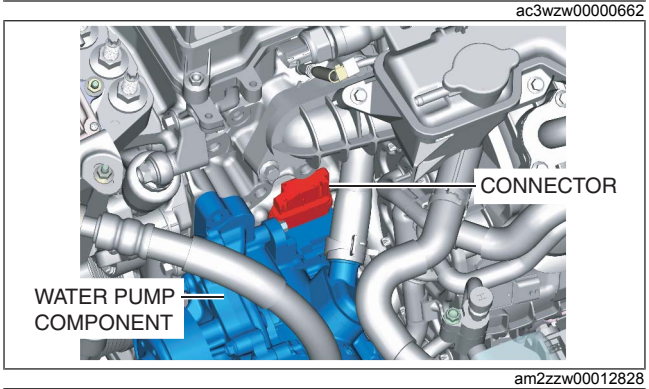


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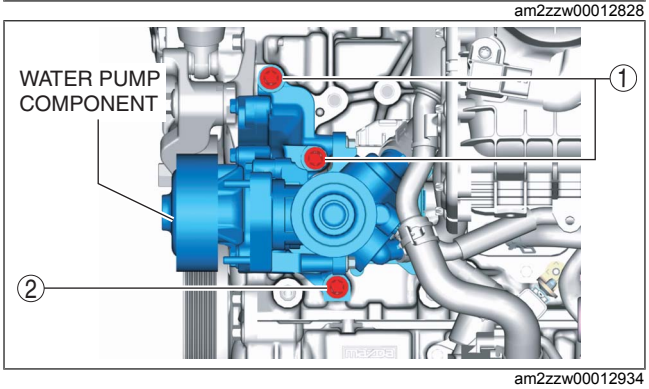
3. Connect the water pump component to the connector.



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4. Tighten the water pump component installation bolts in the order shown in the figure.

Tightening torque
26—32 N·m {2.7—3.2 kgf·m, 20—23 ft·lbf}



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Lower Radiator Hose Component Installation Note

Caution
- **Do not apply oil (such as engine oil, ATF) to the O-ring. Otherwise, the O-ring could swell causing a seal malfunction.**

1. Install the lower radiator hose component to the water pump component from the underside of the vehicle.
2. Connect the lower radiator hose component (radiator side) to the radiator.
3. Tighten the lower radiator hose component installation bolt and nuts.

Tightening torque
8—10 N·m {82—101 kgf·cm, 71—88 in·lbf}